Chapter 1

Demo problem: Solution of a "free-boundary" Poisson problem – a simple model for "fluid"-structure interaction.

In this example we shall consider our first (toy!) interaction problem. The problem combines two single-physics problems, studied in earlier examples, and combines them into a coupled free-boundary problem.

- In one of our first examples we demonstrated the solution of Poisson's equation in a fish-shaped domain, D_{fish}, in which the curvilinear upper and lower boundaries of the fish's body were given by circular arcs which we represented by `GeomObjects`. Given the position of the two circular arcs, whose centres are located at $(X_c, \pm Y_c)$, the single-physics Poisson code computes the solution of Poisson's equation in the corresponding domain.
  We have already demonstrated how `oomph-lib's MacroElement` / `Domain` representation of D_{fish} allows an update of the nodal positions in response to changes in the domain boundary by a simple call to `Mesh::node_update()`. The example code therefore allowed us to compute the solution at a control node, u_{ctrl}, as a function of the "height" of the domain, characterised by Y_c.

- In another example, we demonstrated how to solve a (trivial) solid mechanics problem: The vertical displacement of a circular ring (represented by a `GeomObject`) that is mounted on an elastic foundation of spring stiffness k. The example code allowed us to compute the displacement of the ring, characterised by Y_c, as a function of the load f acting on it.

We will now consider the coupled problem obtained by using the solution of Poisson's equation at the control node, u_{ctrl}, as the "load", f, that acts on the two circular arcs that define the curvilinear boundaries of D_{fish}. The resulting coupled problem is sketched in the figure below. While this problem is obviously somewhat artificial, it has many of the key features that arise in genuine fluid-structure interaction problems. In particular, the displacement of the domain boundary is driven by the solution of the "bulk" (Poisson) equations, just as the deformation of an elastic structure in an FSI problem is driven by the fluid pressure and shear stresses, i.e. quantities that are derived from the solution of the Navier-Stokes equations in the "bulk" domain.

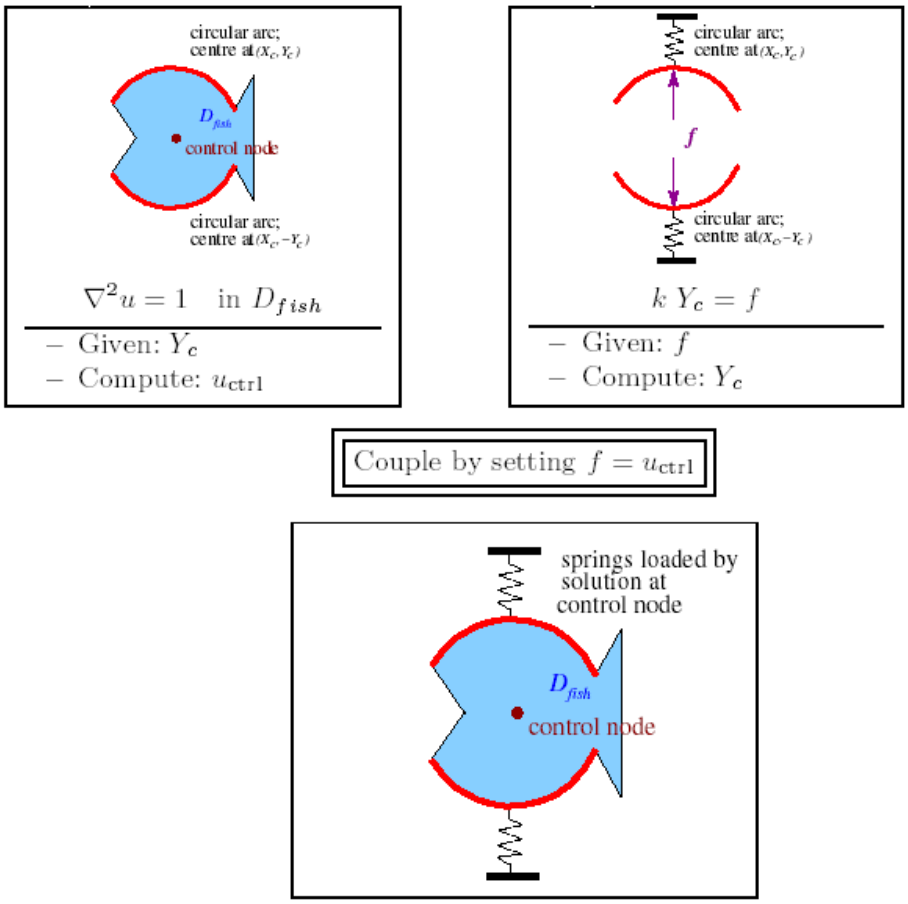

Figure 1.1 Sketch of the two individual single-physics problems (top) and the coupled problem (bottom).

The two single-physics problems involve two uncoupled sets of equations and unknowns:

- The residual vector of the Poisson elements in the single-physics Poisson problem depends on the nodal values in the "bulk" mesh. These nodal values are the only unknowns in the problem since the position of the domain boundary, and hence the position of the nodes are fixed. An update of the nodal positions in response to any changes in the domain boundary is a mere pre-processing step, to be performed just once, before computing the solution.
- The residual vector of the `ElasticallySupportedRingElement` depends on the position of ring's centre, Y_c, which is the only unknown in the problem as the load on the ring is fixed.

The coupling between the two single-physics problem introduces additional dependencies:

- The residuals of the Poisson elements also depend on the nodal positions which in turn depend (via the `MacroElement/Domain` - based node-update function) on the the position of the domain boundary. The boundary position is controlled by the `ElasticallySupportedRingElement's` geometric `Data`, which stores the value of Y_c.
- The residual vector of the `ElasticallySupportedRingElement` also depends on the load, which is now given by the unknown nodal value at a control node in the "bulk" mesh.

We note that most of the methodology required to solve this coupled problem is already available:

- The `MacroElement/Domain` representation of the Mesh makes it possible to update the nodal positions in the bulk mesh in response to changes in the shape/position of the curvilinear domain boundary.

- Multiple inheritance allows the `ElasticallySupportedRingElement` to act as a `GeomObject` (a role in which it can be used to parametrise the unknown curvilinear domain boundary) and as a `GeneralisedElement` (a role in which its unknown geometric `Data` value, Y_c, can be determined as part of the overall solution).
- The load f on the `ElasticallySupportedRingElement` is stored in the element's external `Data`, and derivatives of the element's residual vector with respect to f are automatically taken into account when the element's Jacobian matrix is computed.

The only interaction that still has to be incorporated into the problem formulation is the dependence of the Poisson element's residual vectors on the geometric `Data` in the `ElasticallySupportedRingElement`. This interaction arises through the `MacroElement/Domain` - based node-update function which translates changes in the `GeomObject's` geometric `Data` into changes in the nodal positions. Such dependencies may be added to *any* existing element by "wrapping" the element into the templated wrapper class `MacroElementNodeUpdateElement` which has the following inheritance structure:

```
template<class ELEMENT>
class MacroElementNodeUpdateElement : public virtual ELEMENT,
                                      public virtual MacroElementNodeUpdateElementBase
```

An element of type `MacroElementNodeUpdateElement<ELEMENT>` is an element of type `ELEMENT`, and inherits the additional functionality provided by the `MacroElementNodeUpdateElementBase` base class. The most important additional functionality provided by this class is the ability to add the values stored in the geometric `Data` of associated `GeomObjects` to the element's list of unknowns. Once added, the derivatives of the element's residual vector with respect to these additional unknowns are automatically included into the element's Jacobian matrix. This is achieved by overloading the `ELEMENT::get_jacobian(...)` function and evaluating the additional derivatives by finite differencing. See Comments for details on the implementation.

The solution of the coupled problem therefore only requires a few trivial changes to the single-physics (Poisson) code:

1. The element type used for the solution of the "bulk" equations must be changed to its "wrapped" counterpart, as discussed above. For instance, if the single-physics code used a nine-node refineable Poisson element of type `RefineableQPoissonElement<2,3>`, the coupled problem must be discretised by elements of type `MacroElementNodeUpdateElement<RefineableQPoissonElement<2,3> >` (Yes, it's a bit of a mouthful...).
2. The "bulk" mesh must be "upgraded" (again via multiple inheritance) to a Mesh that is derived from the `MacroElementNodeUpdateMesh` base class.
3. A vector of pointers to those `GeomObjects` that are involved in an element's `MacroElement/Domain` - based node update operation must be passed to the elements. (This is done most easily in the constructor of the "upgraded" mesh.) The geometric `Data` contained in these `GeomObjects` is then automatically included in the elements' list of unknowns.
4. The Mesh's `node_update()` function must be executed whenever the Newton method has changed the values of the unknowns: This is because changing a value that is stored in a `GeomObject's` geometric `Data` does not automatically update the positions of any dependent nodes. This is done most easily be including the `node_update()` function into the `Problem::actions_before_newton_convergence_check()` function; we refer to another document for a more detailed discussion of the order in which the various "action" functions are called by `oomph-lib's` Newton solver.

1.1 Results

The animation below shows the results of a spatially-adaptive solution of Poisson's equations in the fish-shaped domain, for a variety of domain "heights". This animation was produced with the single-physics Poisson solver discussed in an earlier example.

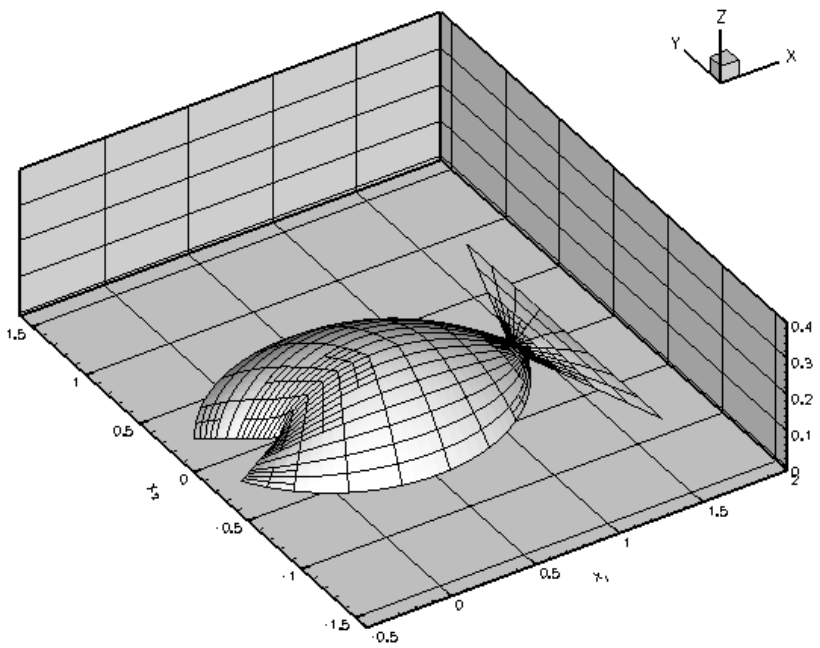

Figure 1.2 Spatially adaptive solution of Poisson's equation in a fish-shaped domain for various `widths' of the domain.

An increase in the height of the domain increases the amplitude of the solution. This is reflected by the red line in the figure below which shows a plot of u_{ctrl} as a function of Y_c. The green marker shows the solution of the coupled problem for a spring stiffness of $k = 1$. For this value of the spring stiffness, the solution of the coupled problem should be (and indeed is) located at the intersection of the curve $u_{ctrl}(Y_c)$ with the diagonal, $u_{ctrl} = Y_c$, shown by the dashed blue line.

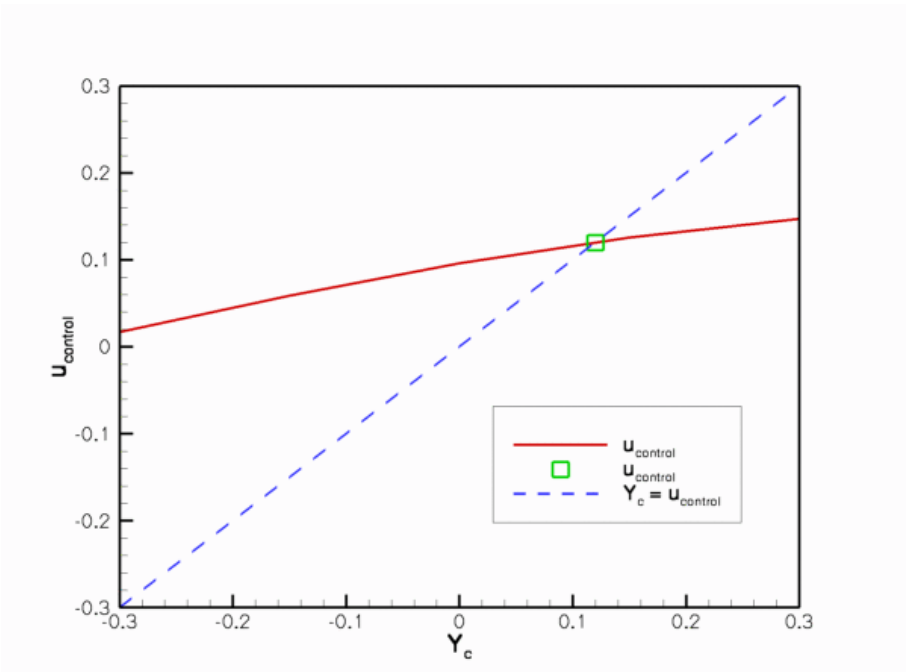

Figure 1.3 Solution of Poisson's equation at a control node as a function of the `height' of the domain.

1.2 Implementation in oomph-lib

The sections below provide the usual annotated listing of the driver code. We stress that only a few trivial changes are required to incorporate the presence of the free boundary into the existing single-physics code:

- The Mesh : Upgrading the `RefineableFishMesh` via multiple inheritance.
- The driver code : Changing the element type for the solution of the Poisson equation.
- The Problem constructor : Storing the element that represents the free boundary in a (sub-)mesh.
- The problem class : Implementing the function `Problem::actions_before_newton_convergence_check()` to update the nodal positions after each Newton step.

1.3 Global parameters and functions

The namespace `ConstSourceForPoisson` defines the constant source function, exactly as in the corresponding single-physics code.

```
//=============start_of_namespace=====================================
/// Namespace for const source term in Poisson equation
//=====================================================================
namespace ConstSourceForPoisson
{

/// Const source function
 void get_source(const Vector<double>& x, double& source)
 {
  source = -1.0;
 }
} // end of namespace
```

1.4 The Mesh

Meshes that are to be used with `MacroElementNodeUpdateElements` should be derived (typically by multiple inheritance) from the `MacroElementNodeUpdateMesh` class. This class overloads the generic `Mesh::node_update()` function and ensures that the node update is performed by calling the `node_update()` function of the `Mesh's` constituent nodes, rather than simply updating their positions, using the `FiniteElement::get_x(...)` function. The overloaded version is not only more efficient but also ensures that any auxiliary node update functions (e.g. functions that update the no-slip condition on a moving fluid node on a solid boundary) are performed too.
In our driver code we add the additional functionality provided by the `MacroElementNodeUpdateMesh` class to the `RefineableFishMesh` class used in the single-physics Poisson problem considered earlier.

```
//==========start_of_mesh==============================================
/// Refineable, fish-shaped mesh with MacroElement-based node update.
//=====================================================================
template<class ELEMENT>
class MyMacroElementNodeUpdateRefineableFishMesh :
 public virtual RefineableFishMesh<ELEMENT>,
 public virtual MacroElementNodeUpdateMesh
{
```

The constructor calls the constructors of the underlying `RefineableFishMesh`. [Note the explicit call to the `FishMesh` constructor prior to calling the constructor of the `RefineableFishMesh`. Without this call, only the default (argument-free) constructor of the `FishMesh` would be called! Consult your favourite C++ book to check on constructors for derived classes if you don't understand this. We recommend Daoqi Yang's brilliant book C++ and Object-Oriented Numeric Computing for Scientists and Engineers.)

```
public:

 /// Constructor: Pass pointer to GeomObject that defines
 /// the fish's back and pointer to timestepper
 /// (defaults to (Steady) default timestepper defined in the Mesh
 /// base class).
 MyMacroElementNodeUpdateRefineableFishMesh(GeomObject* back_pt,
                   TimeStepper* time_stepper_pt=&Mesh::Default_TimeStepper) :
  FishMesh<ELEMENT>(back_pt,time_stepper_pt),
  RefineableFishMesh<ELEMENT>(time_stepper_pt)
```

```
  {
```

To activate the `MacroElementNodeUpdateElement`'s ability to automatically compute the derivatives of the residual vectors with respect to the geometric `Data` that determines its nodal positions, we must pass the pointers to the `GeomObjects` that are involved in the element's `MacroElement` - based node-update to the elements. In general, an element's node-update will be affected by multiple `GeomObjects` therefore the `set_node_update_info(...)` function expects a vector of pointers to `GeomObjects`. In the present example, only a single `GeomObject` (the `GeomObject` that represents the fish's curved "back") determines the nodal position of all elements:

```
   // Set up all the information that's required for MacroElement-based
   // node update: Tell the elements that their geometry depends on the
   // fishback geometric object.
   unsigned n_element = this->nelement();
   for(unsigned i=0;i<n_element;i++)
    {
     // Upcast from FiniteElement to the present element
     ELEMENT *el_pt = dynamic_cast<ELEMENT*>(this->element_pt(i));

     // There's just one GeomObject
     Vector<GeomObject*> geom_object_pt(1);
     geom_object_pt[0] = back_pt;

     // Tell the element which geom objects its macro-element-based
     // node update depends on
     el_pt->set_node_update_info(geom_object_pt);
    }

  } //end of constructor
```

The destructor can remain empty but we provide a final overload for the `Mesh`'s `node_update()` function to avoid any ambiguities as to which one is to be used.

```
 /// Destructor: empty
 virtual ~MyMacroElementNodeUpdateRefineableFishMesh(){}

 /// Resolve mesh update: Node update current nodal
 /// positions via sparse MacroElement-based update.
 //void node_update()
 // {
 //  MacroElementNodeUpdateMesh::node_update();
 // }

}; // end of mesh class
```

1.5 The driver code

The driver code is very simple: We build the problem with the "wrapped" version of the refineable quadrilateral nine-node Poisson element. Since the initial mesh is very coarse we perform two uniform mesh refinements before solving the problem with automatic spatial adaptivity, allowing for up to two further mesh adaptations.

```
//==================start_of_main========================================
/// Driver for "free-boundary" fish poisson solver with adaptation.
//======================================================================
int main()
{

 // Shorthand for element type
 typedef MacroElementNodeUpdateElement<RefineableQPoissonElement<2,3> >
  ELEMENT;

 // Build problem
 FreeBoundaryPoissonProblem<ELEMENT> problem;

 // Do some uniform mesh refinement first
 problem.refine_uniformly();
 problem.refine_uniformly();
 // Solve/doc fully coupled problem, allowing for up to two spatial
 // adaptations.
 unsigned max_solve=2;
 problem.newton_solve(max_solve);
 problem.doc_solution();

} // end of main
```

1.6 The problem class

Apart from a few trivial additions, the problem class is virtually identical to that used in the `single-physics Poisson problem`. The most important addition to the single-physics problem class is the function `Problem::actions_before_newton_convergence_check()` which updates the nodal positions in the "bulk"

Poisson mesh following an update of the geometric `Data` that controls the position of the curvilinear domain boundary; we refer to another document for a more detailed discussion of the order in which the various "action" functions are called by `oomph-lib's` Newton solver.

```
//==========start_of_problem_class=====================================
/// Refineable "free-boundary" Poisson problem in deformable
/// fish-shaped domain. Template parameter identifies the element.
//=====================================================================
template<class ELEMENT>
class FreeBoundaryPoissonProblem : public Problem
{

public:

 ///  Constructor
 FreeBoundaryPoissonProblem();

 /// Destructor (empty)
 virtual ~FreeBoundaryPoissonProblem(){};

 /// Update the problem specs before solve (empty)
 void actions_before_newton_solve() {}

 /// Update the problem specs after solve (empty)
 void actions_after_newton_solve(){}

 /// Access function for the fish mesh
 MyMacroElementNodeUpdateRefineableFishMesh<ELEMENT>* fish_mesh_pt()
  {
   return Fish_mesh_pt;
  }

 /// Doc the solution
 void doc_solution();

 /// Before checking the new residuals in Newton's method
 /// we have to update nodal positions in response to possible
 /// changes in the position of the domain boundary
 void actions_before_newton_convergence_check()
  {
   fish_mesh_pt()->node_update();
  }

private:

 /// Pointer to fish mesh
 MyMacroElementNodeUpdateRefineableFishMesh<ELEMENT>* Fish_mesh_pt;

 /// Pointer to single-element mesh that stores the GeneralisedElement
 /// that represents the fish's back
 Mesh* Fish_back_mesh_pt;

}; // end of problem class
```

1.7 The Problem constructor

We start by creating the `GeomObject/GeneralisedElement` that will represent the unknown curvilinear domain boundary and pass it (in its role as a `GeomObject`) to the constructor of the bulk mesh. We then add the pointer to the bulk mesh to the `Problem's` collection of submeshes and create an error estimator for the adaptive solution of the Poisson equation.

```
//=========start_of_constructor=============================================
/// Constructor for adaptive free-boundary Poisson problem in
/// deformable fish-shaped domain.
//=========================================================================
template<class ELEMENT>
FreeBoundaryPoissonProblem<ELEMENT>::FreeBoundaryPoissonProblem()
{

 // Set coordinates and radius for the circle that will become the fish back
 double x_c=0.5;
 double y_c=0.0;
 double r_back=1.0;

 // Build geometric object that will become the fish back
 ElasticallySupportedRingElement* fish_back_pt=
  new ElasticallySupportedRingElement(x_c,y_c,r_back);

 // Build fish mesh with geometric object that specifies the fish back
 Fish_mesh_pt=new
  MyMacroElementNodeUpdateRefineableFishMesh<ELEMENT>(fish_back_pt);

 // Add the fish mesh to the problem's collection of submeshes:
```

```
 add_sub_mesh(Fish_mesh_pt);

 // Create/set error estimator for the fish mesh
 fish_mesh_pt()->spatial_error_estimator_pt()=new Z2ErrorEstimator;
```

Next we store the pointer to the `ElasticallySupportedRingElement` in its own Mesh and add it to the `Problem's` collection of submeshes before building the `Problem's` global `Mesh` from its two submeshes:

```
 // Build mesh that will store only the geometric wall element
 Fish_back_mesh_pt=new Mesh;

 // So far, the mesh is completely empty. Let's add the
 // GeneralisedElement that represents the shape
 // of the fish's back to it:
 Fish_back_mesh_pt->add_element_pt(fish_back_pt);

 // Add the fish back mesh to the problem's collection of submeshes:
 add_sub_mesh(Fish_back_mesh_pt);

 // Now build global mesh from the submeshes
 build_global_mesh();
```

We choose the central node in the Poisson mesh as the control node and use it (in its role as `Data`) as the "load" for the `ElasticallySupportedRingElement`.

```
 // Choose a control node: We'll use the
 // central node that is shared by all four elements in
 // the base mesh because it exists at all refinement levels.
 // How many nodes does element 0 have?
 unsigned nnod=fish_mesh_pt()->finite_element_pt(0)->nnode();

 // The central node is the last node in element 0:
 Node* control_node_pt=fish_mesh_pt()->finite_element_pt(0)->node_pt(nnod-1);
 // Use the solution (value 0) at the control node as the load
 // that acts on the ring. [Note: Node == Data by inheritance]
 dynamic_cast<ElasticallySupportedRingElement*>(Fish_mesh_pt->fish_back_pt())->
  set_load_pt(control_node_pt);
```

Finally, we pin the nodal values on all boundaries, apply the homogeneous Dirichlet boundary conditions, pass the pointer to the source function to the elements, and set up the equation numbering scheme.

```
 // Set the boundary conditions for this problem: All nodes are
 // free by default -- just pin the ones that have Dirichlet conditions
 // here.  Set homogeneous boundary conditions everywhere
 unsigned num_bound = fish_mesh_pt()->nboundary();
 for(unsigned ibound=0;ibound<num_bound;ibound++)
  {
   unsigned num_nod= fish_mesh_pt()->nboundary_node(ibound);
   for (unsigned inod=0;inod<num_nod;inod++)
    {
     fish_mesh_pt()->boundary_node_pt(ibound,inod)->pin(0);
     fish_mesh_pt()->boundary_node_pt(ibound,inod)->set_value(0,0.0);

    }
  }

 /// Loop over elements and set pointers to source function
 unsigned n_element = fish_mesh_pt()->nelement();
 for(unsigned i=0;i<n_element;i++)
  {
   // Upcast from FiniteElement to the present element
   ELEMENT *el_pt = dynamic_cast<ELEMENT*>(fish_mesh_pt()->element_pt(i));

   //Set the source function pointer
   el_pt->source_fct_pt() = &ConstSourceForPoisson::get_source;
  }

 // Do equation numbering
 cout « "Number of equations: " « assign_eqn_numbers() « std::endl;

} // end of constructor
```

1.8 Post-processing

The post-processing routine writes the computed result to an output file.

```
//============start_of_doc=================================================
/// Doc the solution in tecplot format.
//============================================================================
template<class ELEMENT>
void FreeBoundaryPoissonProblem<ELEMENT>::doc_solution()
{

 // Number of plot points in each coordinate direction.
 unsigned npts=5;

 // Output solution
 ofstream some_file("RESLT/soln0.dat");
 fish_mesh_pt()->output(some_file,npts);
```

```
 some_file.close();

} // end of doc
```

1.9 Comments

A more detailed description of the theory and the implementation can be found in the paper

- Heil, M. & Hazel, A. L. "<TT>oomph-lib</TT> – An <EM>O</EM>bject-<EM>O</EM>riented <EM>M</EM>ulti-<EM>Ph</EM>ysics Finite-Element <EM>Lib</EM>rary". In: *Fluid-Structure Interaction*, Editors: M. Schafer and H.-J. Bungartz. Springer (Lecture Notes on Computational Science and Engineering 53, 2006), (32 pages) (abstract) (pdf preprint).

and in this talk:

- Heil, M. & Hazel, A. L. "An object-oriented approach to the evaluation of the 'shape derivatives' in monolithic fluid-structure interaction solvers". 7th World Congress on Computational Mechanics, LA, July 2006. (pdf).

The following subsections provide a brief description of the main features.

1.9.1 Sparse node updates

The key feature of our implementation which allows the efficient computation of the "shape derivatives" is the ability of `MacroElementNodeUpdateNodes` (discussed in more detail below) to "update their own position" in response to changes in shape/position of the domain boundary. This capability is demonstrated in the following simple example code.

We start by building the Mesh as before

```
//==================start_of_main=========================================
/// Driver to document sparse MacroElement-based node update.
//========================================================================
int main()
{

 // Shorthand for element type
 typedef MacroElementNodeUpdateElement<RefineableQPoissonElement<2,3> >
  ELEMENT;

 // Set coordinates and radius for the circle that will become the fish back
 double x_c=0.5;
 double y_c=-0.2;
 double r_back=1.0;

 // Build geometric object that will become the fish back
  ElasticallySupportedRingElement* Fish_back_pt=
   new ElasticallySupportedRingElement(x_c,y_c,r_back);

 // Build fish mesh with geometric object that specifies the fish back
 MacroElementNodeUpdateRefineableFishMesh<ELEMENT>* Fish_mesh_pt=new
  MacroElementNodeUpdateRefineableFishMesh<ELEMENT>(Fish_back_pt);
```

and document the mesh (i.e. the shape of its constituent finite elements and the nodal positions):

```
 // Number of plot points in each coordinate direction.
 unsigned npts=11;

 ofstream some_file;
 char filename[100];

 // Output initial mesh
 unsigned count=0;
 snprintf(filename, sizeof(filename), "RESLT/soln%i.dat",count);
 some_file.open(filename);
 Fish_mesh_pt->output(some_file,npts);
 some_file.close();
 count++;
```

Next, we "manually" increment Y_c, i.e. the y-coordinate of the centre of the circular arc that defines the upper curvilinear boundary of the fish mesh.

```
 // Increment y_c
 Fish_back_pt->y_c()+=0.2;
```

This step mimics the incrementation of one of the `Problems's` unknowns (recall that in the free-boundary problem considered above, Y_c has to be determined as part of the solution!) during the finite-difference based computation of the shape derivatives.

For meshes that are not derived from the `MacroElementNodeUpdateMeshBase` class, the only way to update the nodal positions in response to a change in the boundary position, is to call the `Mesh::node_update()` function. This updates the position of *all* nodes in the mesh – a very costly operation.

Meshes that are derived from the `MacroElementNodeUpdateMeshBase` class contain `MacroElement↩NodeUpdateNodes` which can update their own position, as shown here:

```
// Adjust each node in turn and doc
unsigned nnod=Fish_mesh_pt->nnode();
for (unsigned i=0;i<nnod;i++)
 {
  // Update individual nodal position
  Fish_mesh_pt->node_pt(i)->node_update();

  // Doc mesh
  snprintf(filename, sizeof(filename), "RESLT/soln%i.dat",count);
  some_file.open(filename);
  Fish_mesh_pt->output(some_file,npts);
  some_file.close();
  count++;
 }

} // end of main
```

We note that the `Node::node_update()` function is defined as an empty virtual function in the `Node` base class, indicating that "normal" `Nodes` cannot "update their own position". The function is overloaded in the `MacroElementNodeUpdateNode` class, details of which are given below. Overloaded versions of this function also exist in various other derived `Node` classes (such as as the `AlgebraicNodes` and the `SpineNodes`) for which algebraic node update operations are defined.
Here is an animation that illustrates how the successive update of the individual nodal positions in response to the change in the boundary position gradually updates the entire mesh.

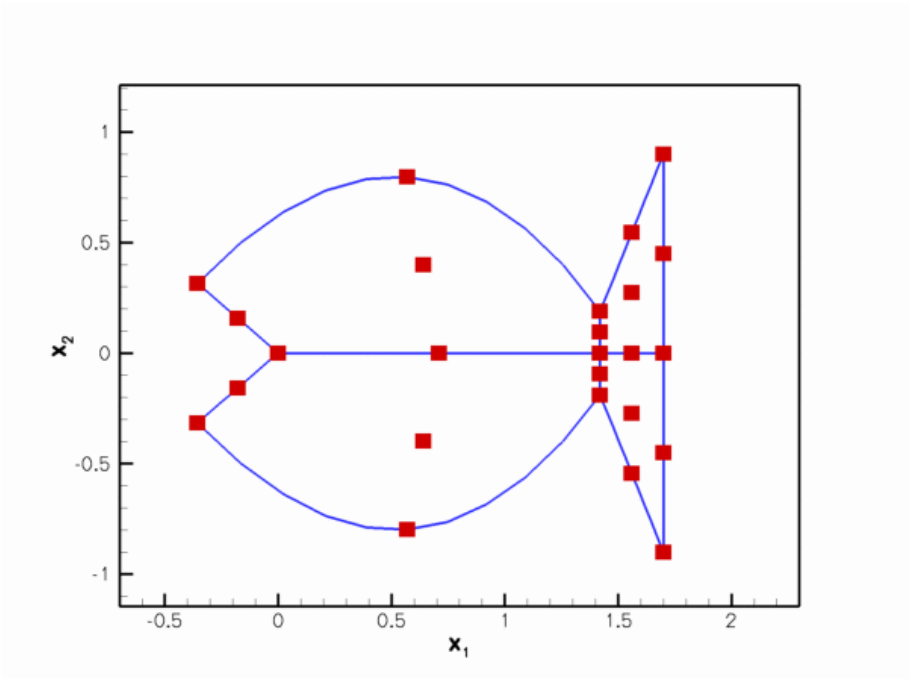

Figure 1.4 Illustration of the sparse node-update procedure.

1.9.2 How it works

The implementation employs three key components:

- **`MacroElementNodeUpdateNodes`** are derived from the `Node` base class. Their main purpose is to provide the `MacroElementNodeUpdateNodes::node_update()` function which updates the nodal position in response to changes in the domain boundary. This capability was demonstrated above and is achieved by allowing the `MacroElementNodeUpdateNodes` to store a pointer to the `Macro↩ElementNodeUpdateElement` that determines its position (using its own `MacroElement` - based representation) and its local coordinates in that element.

MacroElementNodeUpdateNodes also store a function pointer to an auxiliary node update function that allows additional tasks to be performed whenever a node update is performed. This is useful, e.g. in unsteady fluid-structure interaction problems in which a change in the position of nodes that are located on a no-slip boundary also requires an update of the fluid velocities at that node. By default, the function pointer is initialised to NULL, indicating that no auxiliary node update functions have to be executed.

Finally, the MacroElementNodeUpdateNodes store pointers to the GeomObjects that affect their node update. While this information is not required by the node update function itself, it must be available to correctly set up the equation numbering scheme in the presence of hanging nodes. (Details are too messy to explain here but it's true!).

- The **MacroElementNodeUpdateElement<ELEMENT>** class was already discussed in the main part of this document. These elements "wrap around" the element specified by the template argument, ELEMENT, overload some of its member functions and add some new ones.

  Overloaded functions include:

  - The FiniteElement::construct_node(...) functions create an element's local Nodes. This is overloaded by a version that creates MacroElementNodeUpdateNodes instead.

  - The functions GeneralisedElement::get_jacobian(...) and GeneralisedElement::fill_in_contribution_to_jacobian(...) are overloaded by versions that add the shape derivatives to the Jacobian matrices computed by the underlying ELEMENT.

  - Similarly, the function FiniteElement::assign_all_generic_local_eqn_numbers() is overloaded to add the unknowns associated with the node update functions into the element's equation numbering scheme.

  Additional member functions are provided to specify (and access) the GeomObjects that affect an element's MacroElement - based node update. Full details may be found in the "bottom up" discussion of oomph-lib's data structure.

- Finally, the **MacroElementNodeUpdateMeshBase** class overloads the Mesh::node_update() function to ensure that node updates are performed node-by-node, using the MacroElementNodeUpdateNode::node_update() function. This ensures that the node update not only updates the nodal positions but also executes any auxiliary update functions.

1.9.3 The method also works for non-"toy" problems!

The above example demonstrated how easy it is to "upgrade" a driver code for the solution of a single-physics problem to a fluid-structure-interaction-like free-boundary problem. It is important to stress that the methodology employed in our "toy" free-boundary problem can also be used for genuine fluid-structure interaction problems. For instance, the driver code for the simulation of 2D unsteady finite-Reynolds number flow in a channel with an oscillating wall whose motion is prescribed can easily be extended to a driver code for the corresponding fluid-structure interaction problem in which the wall is replaced by a flexible membrane that is loaded by the fluid traction.

1.10 Source files for this tutorial

- The source files for this tutorial are located in the directory:

  `demo_drivers/interaction/free_boundary_poisson/`

- The driver code is:

  `demo_drivers/interaction/free_boundary_poisson/macro_element_free_boundary_poisson.cc`

1.11 PDF file

A `pdf version` of this document is available. \